RegWatch: Registry Change Analysis Report

Generated: 2026-11-03 11:29:03

1. Summary

Metric	Value
Total alerts	3
High severity	3
Medium severity	0
Low severity	0
Info severity	0

2. Integrity Check Results

Metric	Value
Baselines created	1
Clean checks (PASS)	1
Changes logged	3
Baseline tampering detected	0

3. Alerts by Category

Category	Count
Autorun	1
Security	1
Shell	1

4. Alerts by Detection Rule

Rule	Count
Autorun from suspicious location	1
Defender disabled	1
Shell replacement	1

5. Autorun Program Detections

Time	Severity	Value	Program	Message
2026-11-03 11:28:55	High	Updater	C:\Users\Public\fake.exe	New autorun entry detected: C:\Users\Public\fake.exe

6. High-Severity Alerts

Time	Rule	Key	Value	Old	New
2026-11-03 11:28:55	Autorun from suspicious location	HKCU\Software\RegWatchTest\Run	Updater		C:\Users\Public\fake.exe
2026-11-03 11:28:55	Defender disabled	HKCU\Software\RegWatchTest\Security	DisableAntiSpyware		1
2026-11-03 11:28:55	Shell replacement	HKCU\Software\RegWatchTest\Winlogon	Shell		C:\Users\Public\evil.exe

7. Full Change Log

Time	Severity	Type	Key	Value	Old	New
2026-11-03 11:28:55	High	ADDED	HKCU\Software\RegWatchTest\Run	Updater		C:\Users\Public\fake.exe
2026-11-03 11:28:55	High	ADDED	HKCU\Software\RegWatchTest\Security	DisableAntiSpyware		1
2026-11-03 11:28:55	High	ADDED	HKCU\Software\RegWatchTest\Winlogon	Shell		C:\Users\Public\evil.exe

8. Detection Test Results

Metric	Value
Total scenarios	16
True Positives (attacks caught)	9
False Negatives (attacks missed)	2
True Negatives (benign allowed)	5
False Positives (benign flagged High)	0
Accuracy	87.5%
Detection rate (attacks caught)	81.8%
False positive rate	0.0%

Scenario	Type	Expected	Got	Outcome
Autorun from Public folder	attack	High	High	TP
Autorun from AppData with arguments	attack	High	High	TP
Defender disabled	attack	High	High	TP
Real-time protection off	attack	High	High	TP
Firewall disabled	attack	High	High	TP
Shell replaced	attack	High	High	TP
Userinit hijacked	attack	High	High	TP
UAC disabled	attack	High	High	TP
UAC bypass hijack	attack	High	High	TP
Autorun from non-standard folder	attack	High	Medium	FN
Defender: different value name	attack	High	Medium	FN
Normal shell value	benign	NotHigh	Low	TN

UAC left enabled	benign	NotHigh	Low	TN
Defender setting turned on	benign	NotHigh	Medium	TN
Autorun from Program Files	benign	NotHigh	Medium	TN
Legitimate Teams autorun (AppData)	benign	NotHigh	Medium	TN